# High Efficiency Wideband Power Amplifier with Class-J Configuration


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*Abstract*— **The design and implementation of a high efficiency Class-J mode RF power amplifier (PA) for wireless communications applications is described here. In this study, a broadband class-J power amplifier with GaN HEMT transistor is designed to achieve 40 dBm output power and 10 dB gain, high efficiency and linearity performance in the 2.5-3.5 GHz frequency band. Experimental results indicate good agreement with simulations verifying and demonstrating the class-J mode operation. The obtained power added efficiency (PAE) is above 68% and the output power is 39.84 dBm at an input drive of 30 dBm. High efficiency is achieved in the frequency range from 2.5 GHz to 3.5 GHz. Simulation and experimental results show that the class-J RF power amplifier can be realized more effectively in terms of a better compromise between power and efficiency trade-offs over a significant RF bandwidth than conventional class-AB/B operations. the ACPR value for the linearity measurement was found to be around -30 dBc.**



***Keywords— GaN (HEMT), Class-J, high efficiency power amplifier, broadband power amplifier.***


## I. INTRODUCTION

Until recently, in the power amplifier design for wireless communication systems, it focused on a certain low RF bandwidths due to very narrow spectrum allocations. Future emerging technologies like WiMax and 4G, will require larger bandwidths not only because of the need for greater frequency spectrum allocation, but also because of the need to work with signals with baseband bandwidths of 100 MHz and beyond. The results for very high efficiency power amplifiers, typically operating at over 75% efficiency, indicate sensitive multi-harmonic impedance terminations and very high gain-compression levels at the same time. These factors give rise to narrowband frequency performance limitation and nonlinear operation, respectively. As a result, high linearity, efficiency and wide bandwidth are the main important three factors for RF and microwave power amplifiers used in modern communication systems [2].

A demanding for exhibiting high efficiency and linear behaviour in a wide bandwidth causes trade off in classical power amplifiers design. In a class-B power amplifier, the efficiency expectancy is around 75%. This results in a trade-off in the design when attempting to be performed entirely together with other performance criteria parameters. For example, in class-B power amplifiers, if various performance concessions about power and linearity are made, sometimes a efficiency of 78% can be achieved; but a complicated situation arises in terms of providing the other parameters together. This problem have been tried to solve with class-J power amplifiers that have been working on for the last few years. With this new approach, the class-J, the theoretical side of the RF power amplifier design, which does not require a harmonic short circuit that exhibits the same efficiency and linear behaviour of the class-AB mode, is shown [1].

In the class-J mode operation, the phase shift between the output current and voltage waveforms is utilized to transform the second harmonic termination into a fully reactive regime. By this means, a significant contribution is made to the bandwidth-efficiency performance of the class-J designed using GaN (HEMT) power transistors. The class-J power amplifier, terminated with appropriate fundamental and second harmonic impedances, offers a promising solution for efficiency and bandwidth requirements in a single power amplifier design [1,3].

## II. CLASS-J OPERATION MODE THEORY

The class-J power amplifier is introduced by Cripps, which refers to some of Raab's previous works. In class-J operating mode, the output capacitance of the transistor is used effectively for accurate waveform shaping and the desired waveform is obtained at the output. With proper phase shift between output voltage and current waveform, pure reactive load impedance is obtained. This simplifies the design of broadband matching circuits because it does not require an extra open or short resonator circuit. A complex impedance at the fundamental load sustains the drain RF voltage above the "knee" voltage point [1,6].

In the class-J operation mode theory, the second harmonic component is limited to a completely capacitive region to increase the bandwidth of the class-F power amplifier, whereas third and higher harmonics are blocked. Transistor output capacity is used to maintain the fundamental load current and second harmonic current and to not flow higher harmonic currents. The output matching stage allows the fundamental load to be slightly inductive and the second harmonic to be only capacitive [10,13].

In literature about class-J mode, there exists a set of $2^{nd}$ harmonic and fundamental impedance pairs and these impedances form a design space. If the active device is biased as Class-B, all these cases (within the same design region) provide same efficiency and output power like class-B mode. In the class-J power amplifier, only the first two harmonics terms of the class-B drain current shown in Eq.


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(1) are taken. Here, $i_{d,B}$ and $i_{d,J}$ are the class-B and class-J drain current, respectively [1-4].

$$i_{d,B}(\omega t)=I_{maks}\left(\frac{1}{\pi}-\frac{1}{2}\sin(\omega t)-\frac{2}{\pi}\sum_{n=2,4,6,\ldots}^{\infty}\frac{\cos(n\omega t)}{n^2-1}\right) \quad (1)$$

$$i_{d,J}(\omega t)=I_{maks}\left(\frac{1}{\pi}-\frac{1}{2}\sin(\omega t)-\frac{2}{3\pi}\cos(2\omega t)\right) \quad (2)$$

For class-J PA, the fundamental load and harmonic impedance values are as Eq. (3) and Eq. (4), respectively.

$$Z_{f_0,J}=\frac{2(V_{dd}-V_k)}{I_{maks}}(\beta+j\alpha)=\beta R_{opt,B}+j\alpha R_{opt,B} \quad (3)$$

$$Z_{2f_0,J}=-j\frac{3\pi}{8}\frac{2(V_{dd}-V_k)}{I_{maks}}\alpha\beta=-j\frac{3\pi}{8}\alpha\beta R_{opt,B} \quad (4)$$

The α parameter in Eq. (3) and (4) is the reactive component coefficient of class-J and the β parameter is related to the input signal power level. Briefly summarized as the basic operation of class-J, only fundamental and second harmonic terminations are considered in theoretically, as higher harmonics are assumed to be shorted by the output drain-source capacitance [1].

## III. Class-J Amplifier Design

### A. Input and output impedance terminations

Fundamental load-line resistance for the particular operating conditions is determined by simulation. The drain-source capacitance ($C_{ds}$) in reality is non-linear; but $C_{ds}$ capacitance of GaN HEMTs has approximately constant behaviour over $V_{DS}$ DC supply voltage. The input matching is determined through source-pull method [1,3].

The source impedance is determined as the conjugate of the impedance seen from the transistor input; load impedance is obtained together with the fundamental and second harmonic impedances of the load [2]. However, theoretically calculated source and load impedance values in the design are not exactly the same in reality. Therefore, theoretically calculated source and load impedance values are optimized for the most appropriate values in the realization and give different results than their theoretical values. The input and output matching networks are designed to obtain broadband class-J PA using microstrip transmission lines. Fig. 1 shows the design of the class-J power amplifier together with the lumped elements representation of the input and output stages.

### B. Matching Network Design and Layout

With determining values of input and output impedances, the input and output matching stages are designed using a fully transmission lines, except for the input and output decoupling capacitors and a one or two of DC bias bypass capacitors. The second harmonic input and third and above output harmonics are controlled by the matching networks.

The input and output networks are implemented on RT/Duroid 5880 substrate, with $\varepsilon_r$ = 2.2 and thickness of 0.787mm. The designed class-J PA schematic is shown in Fig. 2. The low-loss RT5880 base material used in the high frequency applications of Rogers is used as the base material of the implemented class-J PA circuit. By choosing the dielectric base material with a thickness of 0.254mm, it is possible to obtain much higher characteristic impedance values with sufficiently low line width microstrips. In the realized class-J power amplifier shown in Fig. 2, the element used in the design is the CGH40010F model GaN transistor of Cree. The output power of this transistor is 10 Watt.

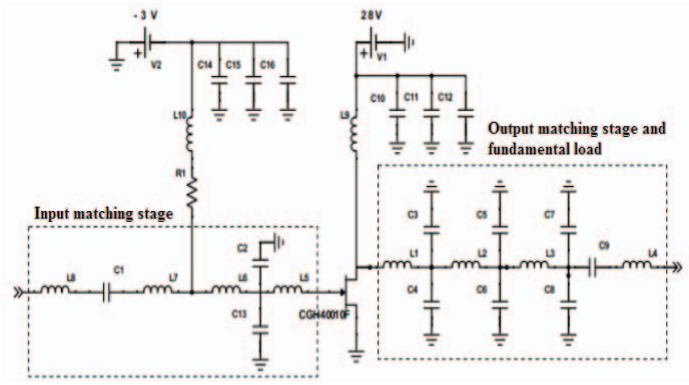



Fig. 1. Schematic of the designed class-J PA with external input and output matching stages with the lumped circuit elements.

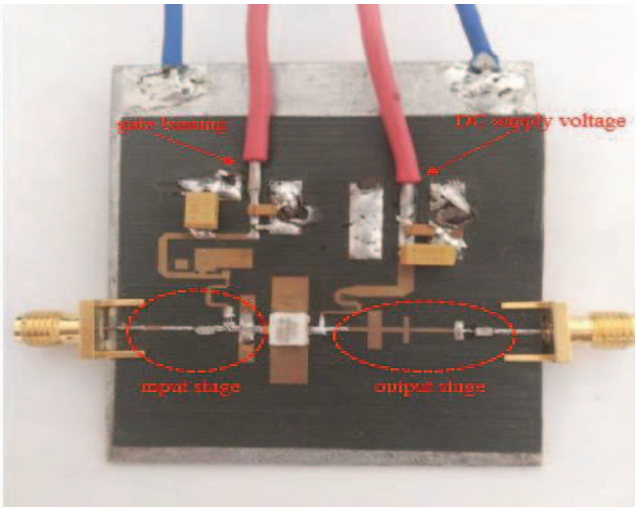



Fig. 2. A photograph of the implemented class-J PA.

## IV. Class-J Performance Measurements

### A. Simulations Results

The electrical operating conditions of the active and passive elements in the PA circuit are determined for simulating and measuring. The gate voltage range is determined from the transistor datasheet for the biasing point. The applied supply voltage in Fig. 2 is 28 V. The value of the voltage to be applied to the transistor gate for biasing in the circuit is set to -3 V. At this value the transistor becomes active. In this way, the transistor is made operational. Fig. 3 shows a graph of the drain current and voltage waveforms and Fig. 4 shows the load voltage waveform obtained from the ADS simulation of the designed class-J power amplifier. These waveforms are similar to expected waveforms in class-J operating mode.



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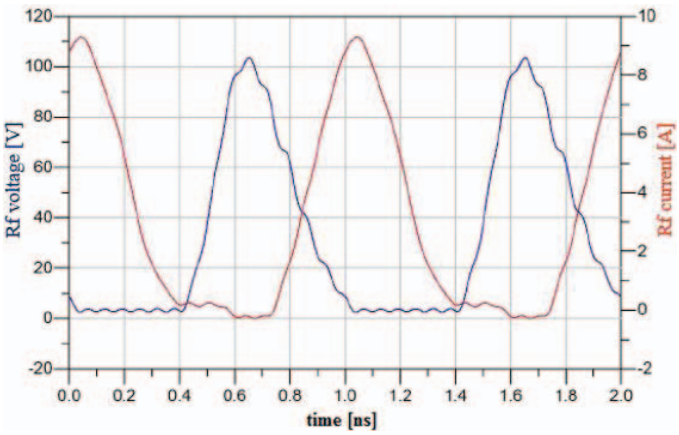



Fig. 3. The RF drain current (red) and the voltage (blue) waveform of the designed class-J PA.

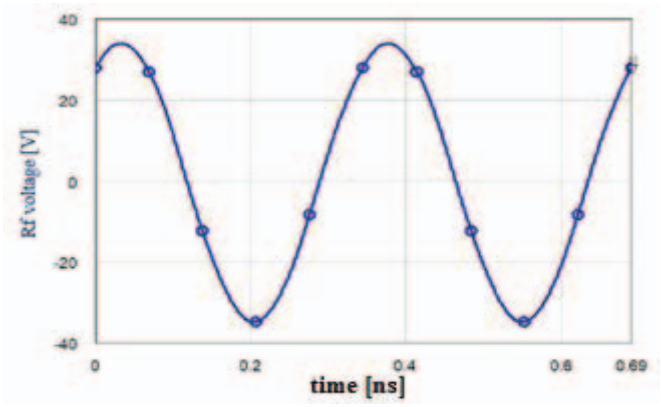



Fig. 4. RF voltage waveform at the load of class-J PA.

Looking over the voltage waveforms in Fig. 3 and Fig. 4, it can be seen that for the input signal with a center frequency of 3 GHz, the voltage on the drain has a half-sinusoidal waveform due to the second harmonic component, while the voltage on the load has a sinusoidal waveform due to matching. Thus, this is an expected condition for the designed class-J power amplifier.

Fig. 5 shows the simulation results of the obtained output power, power added efficiency (PAE) and drain efficiency (DE), when maximum input signal of 30 dBm (1 W) is applied to the designed class-J power amplifier circuit.

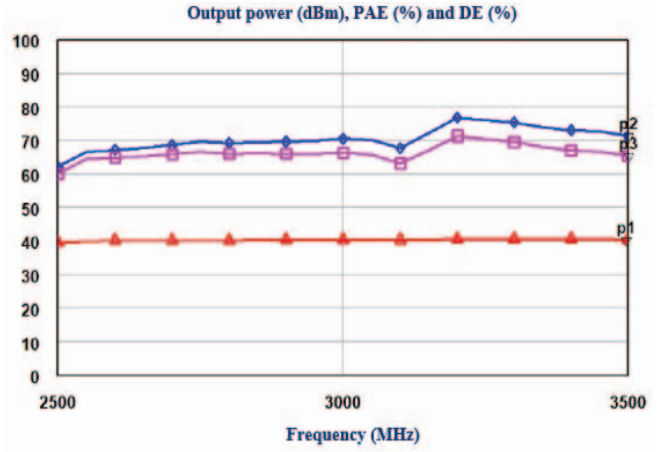



Fig. 5. Output power (red), PAE (pink) and DE (blue) graph depending on the frequency (ADS simulation).

Looking at the simulation graph in Fig. 5, the approximate power added efficiency at the frequency of 3 GHz in the class- J power amplifier is around 69% and the sowing yield is around 71%. The output power obtained in the 2.5-3.5 GHz frequency range is 40 dBm.

In the other test for the linearity simulation, A WCDMA test signal with 5 MHz bandwidth is used to modulate the carrier signal at 3 GHz at the amplifier input. The ACPR value, which is one of the parameters determining the linearity measure of the class-J power amplifier, is simulated.

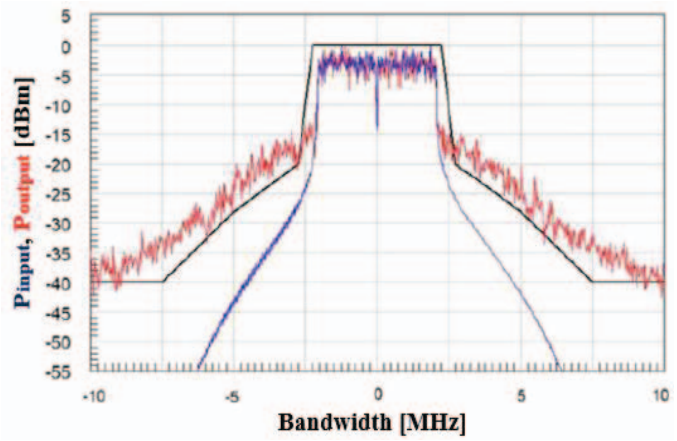



Fig. 6. ACPR graphic obtained at the input (blue) and the output (red) when 5 MHz WCDMA signal is applied, and status of signals with ± 10 MHZ adjacent to the base signal.

### *B. Realized Class-J PA Measurements Results*

The measurement set in Fig. 7 is set up for the measurement of the realized class-J PA. The measured performances under continuous wave input is shown in Fig. 8 for a input signal frequency of 3 GHz.

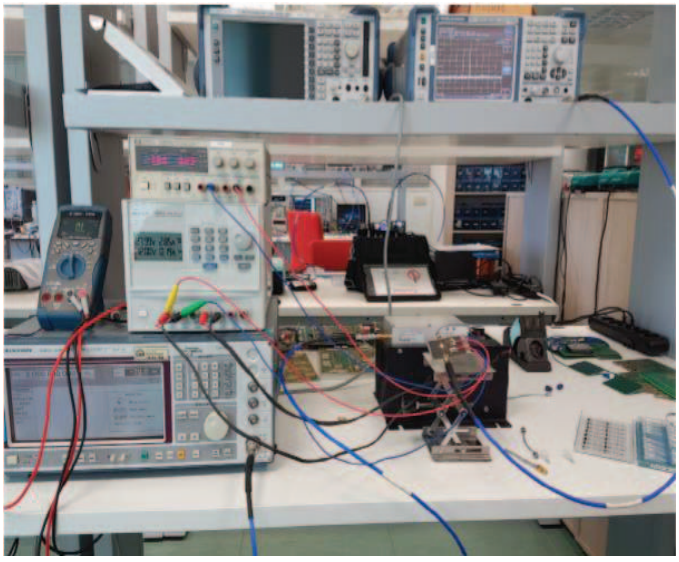


Fig. 7. Measurement setup of class-J PA.



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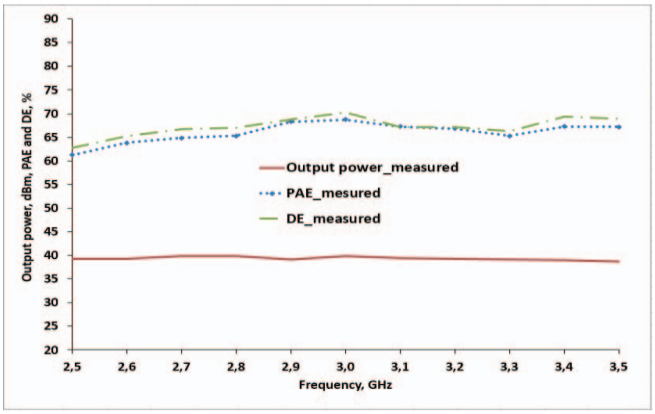



Fig. 8. Measured output power, PAE and DE under CW at Pin of 29.8 dBm.

The simulation results in Fig. 5 and the measurement results in Fig. 8, the output power, the DE and PAE are very close to each other when considering their values.

In a power amplifier, the output power of the fundamental signal at a specific input power point equals the power of the third-order IM components. The input power level at this point is called the third-order input intersection point (IIP3) and the corresponding output power point is called the third-order output intersection point (OIP3). These values are shown in Fig. 9. For the class-J power amplifier realized in this study, the OIP3 value is 49 dBm (for input signal at 3 GHz frequency).

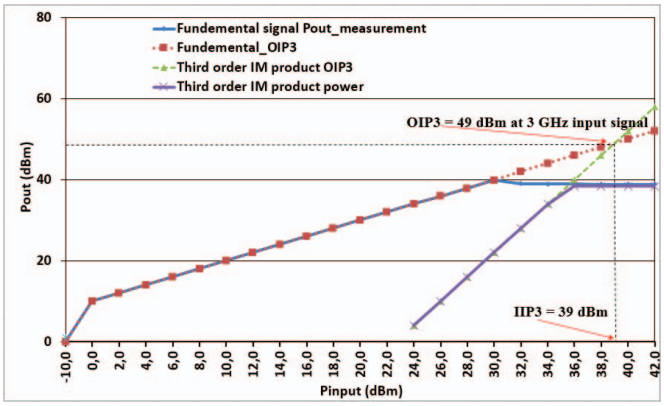



Fig. 9. Measured output powers of the fundamental signal and the third IM products against the input power swept and the showing of the IIP3 and OIP3 points, applying a input signal with 3 GHz frequency.

The performance parameters are summarized to compare the results obtained in this study with similar studies. Table I shows a single-ended class-J PAs performances in recent years.

TABLE I. PERFORMANCE COMPARISON OF THE SOME PUBLISHED CLASS-J PAS

| Ref. | *Bandwidth (GHz)* | *Pout (W)* | *DE (%)* | *PAE (%)* |
|---|---|---|---|---|
| [7] | 1.5-2.5 | 9-11.5 | 60-70 | 61 |
| [11] | 1.6-2.2 | 10-12.6 | 55-68 | 65 |
| [12] | 1.65-2.7 | 9-14 | 55-72 | 69 |
| This work | 2.5-3.5 | 9.8-9.84 | 66-68 | 68 |

## V. CONCLUSIONS

In this study, simulation and realization of the class-J design with a single transistor; broadband, high output and high efficiency power amplifier design has been demonstrated. Class-J power amplifier design combines the linearity of classical linear operation classes with the high efficiency features of linearity and high efficiency power amplifiers. In the realized class-J power amplifier, the output power is 40 dBm in simulation and 39.84 dBm in measurements. The power added efficiency values are 69% and 68% for simulation and measurement, respectively, when the 3 GHz frequency input signal is applied. The gain is fairly flat at around 10 dB in the 2.5-3.4 GHz frequency range. Measurements for linearity evaluation show that the harmonics are at very low levels and the ACPR is around -30 dBc. When the harmonic values and the ACPR value are considered, it is seen that the designed and realized class-J power amplifier exhibits linear operation.

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